

CS228 : Logic for Computer Science

Module 3: Predicate Logic

Lecture 14: Natural Deduction for First-Order Logic

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Welcome back after the Midsem

Module 1: Propositional Logic

1. Syntax
2. Semantics
3. Formal proof systems
4. Normal Forms

Module 2: The Problem of Satisfiability

1. Propositional Resolution
2. DPLL
3. CDCL
4. SAT/SMT tool demo

Module 3: Predicate Logic

1. Syntax of predicate logic (Lecture 12)
2. Semantics of predicate logic (Lecture 13)
3. Today: a formal proof system for predicate logic (natural deduction)

Let's warm up with a quick recap before diving into today's new material.

Predicate Logic: Alphabet and Syntax

Alphabet consists of

- ▶ Variable names: x, y, z
- ▶ **Vocabulary**
 - ▶ Function symbols: $f, g, h \dots$
 - ▶ Constants: a, b, c are just 0-ary function symbols.
 - ▶ Relation (Predicate) symbols: $P, Q, R, \dots$
- ▶ Fixed symbols: $\wedge, \vee, \neg, \rightarrow, (,), \exists, \forall$

Syntax of First Order Logic (quantification over variables)

Two classes of syntactic objects: **terms** and **formulas** defined over the alphabet.

- ▶ Terms
$$t ::= x \mid c \mid f(t, \dots, t)$$

where x is over variables, c over constants and f over other function symbols.

- ▶ Formulas
$$\varphi ::= P(t_1, \dots, t_n) \mid (\neg \varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi) \mid (\forall x \varphi) \mid (\exists x \varphi)$$

where P is over set of predicates, $t_1, \dots, t_n$ are terms and x is a variable.

Substitution in FOL

Variables are placeholders; so we must be able to replace them with concrete terms: substitution.

- ▶ Suppose $x \in \text{free}(\varphi)$ and t is any term.
- ▶ For example, consider $\psi = \exists y R(x, y) \vee \forall x R(z, x)$.

Definition

Term t is free for x in φ if no free occurrence of x in φ is in the “scope” of $\forall y$ or $\exists y$ for any var y occurring in t . e.g., $f(z, x)$ is free for x in ψ , but $f(y, x)$ is not.

Substitution

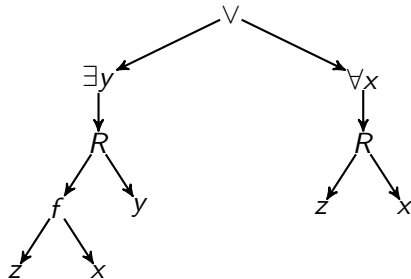
$\varphi[t/x]$ is formula obtained by replacing each free occurrence of x in φ by t , if t is free for x in φ .

For ψ defined above: $\psi[f(z, x)/x] := \exists y R(f(z, x), y) \vee \forall x R(z, x)$

Substitution in FOL: Correct vs. Captured

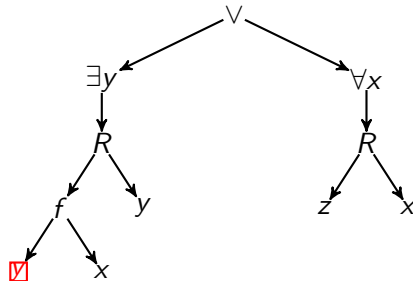
Correct: $\psi[f(z, x)/x]$

$\exists y R(f(z, x), y) \vee \forall x R(z, x)$



Wrong: $\psi[f(y, x)/x]$

$\exists y R(f(y, x), y) \vee \forall x R(z, x)$



The y inside $f(y, x)$ (boxed in red) is now captured by $\exists y$ – it's a different leaf than the term's original free y , but shares its scope!

- **Fix:** rename the bound y to a fresh y' first
- Rewrite ψ as $\exists y' R(x, y') \vee \forall x R(z, x)$ (same meaning, y is just a placeholder), *then* substitute.

Quick Recap: Semantics of Predicate Logic

- ▶ **Vocabulary** $\mathcal{V}$ + **universe** U + interpretation of constants/functions/predicates on $U \Rightarrow$ a $\mathcal{V}$ -structure (model) M
- ▶ **Environment** $\alpha : \text{var}(\varphi) \rightarrow U$ (defined on all variables, but only its values on **free** variables affect satisfaction)
- ▶ **Satisfaction relation** $M, \alpha \models \varphi$ defined by structural induction on φ (atomic formulas via interpretation; $\wedge, \neg$ as usual; $\exists x \varphi$ iff some $u \in U$ with $M, \alpha[x \mapsto u] \models \varphi$; similarly for $\forall$)
- ▶ For a **sentence** (no free variables), we just write $M \models \varphi$ – the environment doesn't matter
- ▶ **Semantic entailment** $\Gamma \models \psi$, **validity**, and **satisfiability** are defined just as in propositional logic, with M, α in place of a truth assignment

Unlike propositional logic, both **validity** and **satisfiability** for FOL are undecidable – no algorithm can always decide them in finite time.

Today: can we nevertheless *prove* things about FOL formulas syntactically?

Topic 14.1

Natural Deduction for FOL

A Proof System for FOL

- ▶ All proof rules for **Propositional Logic** carry over and remain sound for FOL.
- ▶ In addition, we need new rules for:
 - ▶ **Equality** ($=$)
 - ▶ **Quantifiers** ($\forall, \exists$)

Equality Rules

- ▶ A rose is a rose!
- ▶ If two terms are equal, then we can replace one by the other in any formula.
- ▶ If $y * (w + 2) = y * w + y * 2$; then $z \geq y * (w + 2)$ implies $z \geq y * w + y * 2$ (and vice versa)

$$\frac{}{t = t} =_i \qquad \frac{t_1 = t_2 \quad \varphi[t_1/x]}{\varphi[t_2/x]} =_e$$

- ▶ Axiom for equality, no premises
- ▶ Only on terms NOT formulas
- ▶ Of course, this is applicable ONLY if t_1, t_2 are free for x in φ .
- ▶ Eliminates $=$ by replacing all t_1 in $\varphi[t_1/x]$ by t_2 .

Exercise 14.1 Show that the following sequents are valid.

1. $t_1 = t_2 \vdash t_2 = t_1$
2. $t_1 = t_2, t_2 = t_3 \vdash t_1 = t_3$

Exercise 14.1: Solution

1. $t_1 = t_2 \vdash t_2 = t_1$

1. $t_1 = t_2$ premise
2. $t_1 = t_1$ $=_i$
3. $t_2 = t_1$ $=_e$ 1,2

where φ is $x = t_1$, so line 2 is $\varphi[t_1/x]$ and line 3 is $\varphi[t_2/x]$.

2. $t_1 = t_2, t_2 = t_3 \vdash t_1 = t_3$

1. $t_1 = t_2$ premise
2. $t_2 = t_3$ premise
3. $t_1 = t_3$ $=_e$ 2,1

where φ is $t_1 = x$, so line 1 is $\varphi[t_2/x]$ and line 3 is $\varphi[t_3/x]$.

Proof Rules for Universal Quantification

$$\frac{\forall x \varphi}{\varphi[t/x]} \forall x e$$

$$\frac{\boxed{\begin{array}{c} x_0 \\ \vdots \\ \varphi[x_0/x] \end{array}}}{\forall x \varphi} \forall x i$$

- ▶ All students did PS1 $\Rightarrow$ so did YOU!
- ▶ If $\forall x \varphi$ holds, then we can replace x in φ by any term t .
- ▶ **Imp:** side condition t is free for x in φ .
 - ▶ Consider $\varphi = \exists y(x < y)$ with vocab reals with usual smaller than relation.
 - ▶ $\forall x \varphi$ means for all real n , there exists real m bigger than n , **true!**
 - ▶ But $\varphi[y/x] = \exists y(y < y)$, which says there must exist a no. bigger than itself, **false!**
 - ▶ We want only sound rules!
 - ▶ Soln: replace variable y by fresh var z
- ▶ Say I threw this tennis ball $\Rightarrow$ I can throw ANY tennis ball!
- ▶ If from a fresh var x_0 , we can prove $\varphi[x_0/x]$, then $\forall x \varphi$ holds.
- ▶ Side condn: x_0 does **not** occur outside the box.
- ▶ In proving φ , we didn't make an assumption on x .

Examples

Exercise 14.2 Prove that the following sequents are valid.

1. $\forall x(P(x) \rightarrow Q(x)), \forall xP(x) \vdash \forall xQ(x)$
2. $P(t), \forall x(P(x) \rightarrow \neg Q(x)) \vdash \neg Q(t)$ for any term t .

Exercise 14.2: Solution

1. $\forall x(P(x) \rightarrow Q(x)), \forall xP(x) \vdash \forall xQ(x)$

1. $\forall x(P(x) \rightarrow Q(x))$ premise

2. $\forall xP(x)$ premise

3.	$P(x_0) \rightarrow Q(x_0)$	$\forall x e 1$
4.	$P(x_0)$	$\forall x e 2$
5.	$Q(x_0)$	MP 3,4

6. $\forall xQ(x)$ $\forall x i 3-5$

2. $P(t), \forall x(P(x) \rightarrow \neg Q(x)) \vdash \neg Q(t)$

1. $P(t)$ premise

2. $\forall x(P(x) \rightarrow \neg Q(x))$ premise

3. $P(t) \rightarrow \neg Q(t)$ $\forall x e 2$

4. $\neg Q(t)$ MP 3,1

(box scoped by fresh x_0)

An Interesting Analogy

Intuition: $\forall$ “generalizes” $\wedge$

- ▶ Just like in $\wedge$ -e, we could get F or G from $F \wedge G$, we can get $\varphi[t/x]$ for any term t (modulo side-condn!).
- ▶ Similarly $\wedge$ -i had two premises, here $\forall x \ i$ has premise for any x_0 .

Proof rules for Existential Quantifiers

- ▶ Can we lift this analogy and do the same for $\exists$ and $\forall$?
- ▶ For $\forall$, $\forall$ -i was easy! So what should be the rule for $\exists$ -i? What about $\exists$ -e?

$$\frac{\varphi[t/x]}{\exists x \varphi} \exists x i$$

$$\frac{\exists x \varphi \quad \boxed{\begin{array}{c} x_0 \quad \varphi[x_0/x] \\ \vdots \\ \chi \end{array}}}{\chi} \exists x e$$

- ▶ To use $\exists x \varphi$, you should have a proof from any possible value for x .
- ▶ The box controls scope of x_0 and scope of assumption also.

Examples

Exercise 14.3 Show that the following sequents are valid.

1. $\forall x\varphi \vdash \exists x\varphi$
2. $\forall x(P(x) \rightarrow Q(x)), \exists xP(x) \vdash \exists xQ(x)$
3. $\exists xP(x), \forall x\forall y(P(x) \rightarrow Q(y)) \vdash \forall yQ(y)$

Note: Be very careful with scope of $x_0 \dots$ see examples in Huth & Ryan for “false” proofs that one may derive by misusing the scope.

Many more examples in Huth & Ryan... Do try them out!

Solution for 1. $\forall x\varphi \vdash \exists x\varphi$

1. $\forall x\varphi$ premise
2. φ $\forall x \text{ e } 1$ (choose $t = x$ for both steps: $\varphi[x/x] = \varphi$, so this is legal)
3. $\exists x\varphi$ $\exists x \text{ i } 2$

Exercise 14.3.2: Correct vs. Wrong

Correct

1.	$\forall x(P(x) \rightarrow Q(x))$	premise
2.	$\exists xP(x)$	premise
3.	$P(x_0)$	assumption
4.	$P(x_0) \rightarrow Q(x_0)$	$\forall x e 1$
5.	$Q(x_0)$	MP 4,3
6.	$\exists xQ(x)$	$\exists x i 5$
7.	$\exists xQ(x)$	$\exists x e 2, 3-6$

Wrong!

1.	$\forall x(P(x) \rightarrow Q(x))$	premise
2.	$\exists xP(x)$	premise
3.	$P(x_0)$	assumption
4.	$P(x_0) \rightarrow Q(x_0)$	$\forall x e 1$
5.	$Q(x_0)$	MP 4,3
6.	$Q(x_0)$	$\exists x e 2, 3-5$
7.	$\exists xQ(x)$	$\exists x i 6$

The box on the right closes one line too early: line 6 ($Q(x_0)$) lets the fresh x_0 escape its own box before $\exists x i$ is applied outside – but x in $\exists x e$ must not mention x_0 !